

Smart IoT Device Troubleshooting Chatbot Using Agentic AI

Modern IoT environments consist of numerous interconnected devices such as sensors, smart appliances, industrial equipment, and embedded systems. When these devices experience issues such as connectivity failures, sensor malfunctions, firmware errors, battery problems, or configuration errors, troubleshooting often requires technical expertise and significant manual effort. Traditional support systems rely on static FAQs, documentation, or human technicians, which can result in delayed issue resolution, increased operational costs, and prolonged device downtime. There is a need for an intelligent, autonomous, and context-aware troubleshooting assistant capable of diagnosing IoT device issues, suggesting corrective actions, and guiding users through the resolution process in real time.

The Challenges

1. Diverse IoT Devices like Different manufacturers Different communication protocols, Different error formats
2. Real-Time Diagnostics Needed like for immediate analysis, Handling streaming device data
3. Knowledge Retrieval- Accessing large troubleshooting knowledge bases, Retrieving relevant solutions accurately
4. Multi-Step Reasoning- Diagnosing complex failures often requires multiple steps, Agent coordination is required
5. Data Integration- Integrating logs, sensors, and device status information
6. Context Management- Maintaining conversation history, Understanding previous troubleshooting attempts
7. Scalability- Supporting thousands of IoT devices simultaneously
8. Explainability- Providing understandable reasoning behind recommendations
9. Security and Privacy- Protecting device information, Secure communication between chatbot and IoT systems
10. Handling Uncertainty- Incomplete logs, Missing sensor data, Ambiguous user descriptions

The Objectives

- Develop an intelligent Agentic AI chatbot that can automatically diagnose and troubleshoot IoT device issues through natural language interaction.
- Automate IoT device troubleshooting.
- Provide real-time troubleshooting guidance.
- Reduce device downtime.
- Improve user experience.
- Enable multi-step reasoning using AI agents.
- Integrate IBM Granite Models for intelligent diagnosis.
- Use LangFlow to orchestrate agent workflows.
- Generate detailed troubleshooting reports.
- Support multiple IoT device categories.
- Reduce dependency on human support teams.

Tech Stack

- LangFlow + IBM watsonx.ai + Granite Models